

Epigenetic changes due to physical activity

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Abstract

One of the epigenetic-modifying factors is regular and continuous physical activity. This article attempts to investigate the effects of physical activity and exercise on changes in histone proteins and gene expression, as well as the effect of these exercises on the prevention of certain cancers and the ejection of age-related illnesses and cellular oxidation interactions. All of this is due to epigenetic changes

and gene expression. Most studies have reported the positive effects of regular exercises on the expression of histone proteins. DNA methylation and the prevention of certain diseases such as cancer and respiratory diseases, caused by antioxidative interactions that occur more often in the elderly, have been studied. © 2018 International Union of Biochemistry and Molecular Biology, Inc. Volume 65, Number 6, Pages 761–767, 2018

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1. Introduction

The epigenetic mechanisms change the chromatin structure by changing the components of the chromatin. This, in turn, changes the pattern of gene expression. The most important epigenetic mechanisms include chromatin-modifying factors, histone-modifying agents and histone tubes, DNA methylation, and small regulatory RNAs (miRNAs). Chromatin is a collection of DNA and DNA-linked proteins, including histones and nonhistone chromosomal proteins. At the beginning of the discovery of the histones, they were given different names, but today their naming is as follows: H1, H2A, H2B, H3, and H4 [1].

Histone proteins play a special role in the chromatin binding, and as nucleus-forming units they play a very important

role in shaping the structure of chromatin and gene expression. Histone proteins are exposed to a series of posttranslational changes, and trunks of these histone modifications and changes can occur throughout the amino acid sequence of the histones, but often the nonstructural ends, that is, histones are involved. These changes include acetylation, methylation, deacetylation, phosphorylation, and ubiquitination of amino acids forming histone proteins. In fact, these changes preserve epigenetic information and affect the transcription of the gene and DNA changes [2].

Studies have shown that exercise activity causes changes in histones and changes in the cell nucleus by factors such as Ila and HDAC, as well as phosphorylation by CAMKII or active protein kinase AMP (AMPK), which will all increase and decrease the gene expression after exercise. The unveiling of changes in histone proteins and DNA changes, especially before and after exercise, is challenging, but exercise as an effective factor in altering these proteins, along with DNA changes, including DNA methylation and in general, changes in gene expression, detailed studies in this field will provide a better understanding of the regulation of gene expression during exercise and after exercise.

On the other hand, these current conditions cause a series of changes in skeletal muscle and also the metabolic process in both healthy people and patients [3, 4]. In addition, the effect of these changes will be on the process of preventing and improving age-related illness and improving the quality of

Abbreviations: CaMKI, II, IV, protein kinase 1, 2, and 4; HAT, histone acetyltransferase; HKII, hexokinase II; HMTs, methyl-transferase; MCAD, dehydrogenase; Mgn, myogenin; PDK4, isoenzyme pyruvate dehydrogenase kinase 4; PGC-1 α , alpha-coactivator; PKD, protein kinase D; PPAR- γ or PPARG, activated receptor gamma; SIK1, induced salt kinase; TFAM, mitochondrial transcription factor.

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TABLE 1

Histone methylation sites (adapted from Ref. [14] with permission)

Histone	Position	Modification	Transcriptional effect/gene or chromatin location
H3	K4	Di-methyl	Gene activation
		Tri-methyl	Gene activation/5' end transcriptionally active gene
	K9	Mono-methyl	Gene silencing/euchromatin
		Di-methyl	Gene silencing/euchromatin
		Tri-methyl	Gene silencing/promoters and heterochromatin
H4	K27	Tri-methyl	Gene activation/gene coding region
	K79	Mono-methyl	Gene silencing/heterochromatin
		Tri-methyl	Gene silencing/inactive X-chromosome, imprinted regions, and homeotic genes
	K20	Di-methyl	Gene activation
		Tri-methyl	Gene activation

life and delaying the aging process [5]. For this reason, in the present article, we aimed to review the effects of exercise and regular physical activity on epigenetic changes, and the effects of these changes on improving the health and life.

2. Epigenetic markers

Epigenetic changes, such as changes in transcription of histone proteins [6], play a key role in gene expression and changes in the structure of chromatin [7, 8]. By recognizing the properties of histone proteins, it has been shown that differences in size, amino acids, especially lysine and arginine, and finally solubility are different. Acetylation, deacetylation, and methylation of histones, especially in H3 and H4 histones (Table 1), have been most studied among histone changes. Histone acetylation by the enhancement of histone acetyltransferase (HAT) enzymes will increase digestion, and their deacetylation by HDACs in most cases will reduce the number of copies [3, 9]. Meanwhile, methylation by the methyl-transferase (HMTs) enzyme can, depending on the type and position of the methylated amino acids, also disable the duplication and also activate the transcription in some genes [10, 11].

Evidence is increasing [12] that exercise can intensely change the epigenome (e.g., via the methylation of DNA) for

the purpose of short-term regulation (e.g., the expression of mRNA). More theoretically, possibly, maternal exercise at the time of pre- and postpartum could lead to epigenetic changes in offspring. It is normally understood that there are importance of consistent mild exercise during pregnancy [13]. The phenotypic advantages of maternal exercise, nonetheless, can be perceived as a type of organismal stressor.

There is likewise an increase in literature on the role of small noncoding RNAs (sncRNAs) and their consequences on transcriptional gene silencing. These RNAs comprise the small inhibitory RNAs (siRNAs) synthesized by the RNA-interference pathways; the dicer-dependent microRNAs (miRNAs); the piRNAs, synthesized in a dicer-independent fashion; and especially related with the PIWI subfamily of argonaute proteins. While each of these groups of sncRNAs have been implicated in epigenetic DNA and histone modifications, the piRNAs shows to have a different role of inhibiting transposon expression in the cells of the germline through fostering *de novo* methylation of DNA [15]. The role of siRNA in RNA-mediated methylation of DNA and gene transcription silencing was first discovered in plants [16] and since then, it has been found to be present in several species, which include but not limited to mammals [17].

2.1. Exercise and muscle structure

Changes in histones and their effect on transcription of various genes are one of the challenges of modern biomolecular science [2]. HDAC5 and Ila appear to play a role in regulating gene expression and increasing the volume of muscle fibers and their shape. In fact, by increasing the expression of the oxidative gene by alterations of HDAC and Ila, it causes some changes in the expression of carnitine transferase I, hexokinase II (HKII), the medium-chain acyl-CoA dehydrogenase (MCAD), ATP synthase- β , glycogen phosphorylation, synthesis and an increase of many oxidative genes in skeletal muscle [18].

Studies have shown that exercise is one of the factors behind the creation of these processes. For example, Mahoney et al. found that regular physical activity would modify histone proteins and lead to the expression of some genes [19]. Another study by Harcourse et al. found that immediately after exercise, the acetylation would increase the histone H3 protein of skeletal muscle [20]. In human studies, it has been shown that HDAC5 and Ila can play a very important role in controlling the gene expression after exercise [21]. Sun et al. 2009, in a study on human skeletal muscle, found that aerobic exercise caused changes to HDAC and Ila [22].

In the past, it was believed that only calcium-dependent mechanisms would increase or decrease the expression of the gene, but further studies showed that changes in histone proteins by using calcium-dependent mechanisms will increase the expression of oxidative genes, kinases, by mediating changes in histone proteins, including calcium-calmodulin kinase-dependent proteins, protein kinase 1, 2, and 4 (CaMKI, II, IV) [1, 23], protein kinase D [5, 6], active protein kinase AMP (AMPK) [24], and AMPK kinases associated with induced salt kinase (SIK1) [4]. Exercise is one of the known factors for

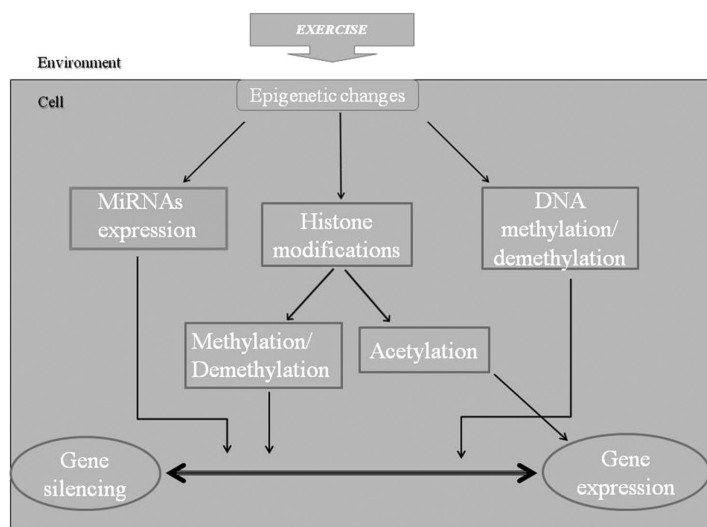


FIG. 1

A review of epigenetic changes induced by exercise. This figure adapted from [6] with copyright permission.

activating some of these kinases in skeletal muscle. Some of these kinases have been identified as agents for increasing or decreasing the process I α and HDACS [4].

Another process that influences the changes of I α and HDACS is the mechanism that is mediated through ubiquitin. This process involves the E3 of ubiquitin and the binding of the ubiquitous peptide to the HDAC [21]. However, this mechanism has not yet been studied extensively and has not yet been proven to be accurate [21]. In some studies, it has also been shown that with the disruption of the nervous system, the nerve agent, HDAC will increase and alter the gene expression [25, 26]. These studies have shown that the expression of two factors, HDAC and CAMKII, can act as an important and influential factor in neuromuscular activity [26]. The repression of HDAC activity attenuates activity-regulated genes induction in adult denervated muscle and aneural myotubes. The effect of the inhibitors of HDAC necessitates new protein synthesis, suggesting that HDACs may control the expression of a myogenin (Mgn) transcriptional repressor [27].

As shown in Fig. 1, even a short period of regular physical activity can cause genome hypermethylation within the muscle cells. This means that many regulatory genes can be transformed into pathways such as regeneration and muscle growth. Exercise intensity is directly related to the amount of promoter demethylation, so intense physical activity activates more genes [6]. Other factors related to the mentioned cases, such as mRNA and MiRNAs, regulate many physiological processes, such as the prevention of inflammation, dehydration (the formation of blood vessels), as well as the prevention of ischemia (restriction of blood flow in the vessels). miRNAs are a family of small RNAs with an approximate size of 21–25 nucleotides. These molecules control the expression of specific genes under controlled transcription, thereby regulating gene

expression [2]. Aerobic exercise activates the total number of different miRNAs within the skeletal muscle cell, which reduces the negative effects of the stimulation (Fig. 1) [6].

In addition, exercise through muscle growth and aerobic endurance can have widespread effects on diabetes and other metabolic diseases [6]. Increasing the number of type I muscle fibers that are reinforced by aerobic activity will lead to an increase in glucose and lipid metabolism and an important consequence of the prevention and treatment of many metabolic diseases [21]. Promoter methylation and activation of genes depend heavily on exercise. Extreme moderate and relatively high exercise causes reduced methylation of active promoters such as peroxisome proliferator-activated receptor gamma, alpha-coactivator (PGC-1 α), mitochondrial transcription factor (TFAM), isoenzyme pyruvate dehydrogenase kinase 4, and MEF2A, immediately after physical activity as well as a delayed methylation of active promoters receptors (PPAR δ) 3 H after exercise. This exercise can cause cytosolic changes in humans and animal samples such as mice. These changes can be due to changes in Ca²⁺ and AMP. Both of which are involved in the activation of signaling cascades [6, 28]. Ca²⁺/calmodulin, AMP, protein kinase (CaMK), and kinase-dependent protein (AMPK) via HDACs alter the expression of the gene in humans and cause the expression of the gene, including the transfer genes glucose 4 (GLUT4), a potent inhibitor of diabetes in healthy people and people with type 2 diabetes mellitus (Fig. 2) [6, 28].

2.2. Exercise and cancer

Exercise leads to epigenetic changes that can have beneficial effects in cancer patients. The effect of exercise on DNA methylation patterns leads to increased expression of the gene associated with tumor suppression and decreased expression of oncogenes [29, 30].

DNA methylation involves the addition of an open-minded mitochondria of cytogenes in regions rich in CPG dyne nucleotide that are known as CPG. The cytosine methylation is created after the formation of DNA, and as a result of the transfer of the enzymatic transfer of a mitochondrial (-CH₃) group of -S adenosine methionine to the 5-carbon cytosine present in the CPG dinucleotide.

This enzymatic reaction is performed by the DNA methyl transferase [23]. Cancer cells have abnormal DNA methylation patterns, including hypermethylation in tumor suppressor gene promoter regions and hypomethylation in promoter regions of oncogenes [29].

This epigenetic alternation in cancer cells will cause the cell to grow and break down, resulting in a tumor. Exercise to reduce this mutation and even reverse epigenetics has been shown to increase the level of expression of the tumor suppressor gene and reduce the level of oncogenesis [29, 30]. As illustrated in Fig. 3, the broad DNA methylation in adults, which is routinely >30 Min of regular exercise per day, is significantly increased compared to those with low mobility, which helps to prevent and proliferate cancer cells [31].

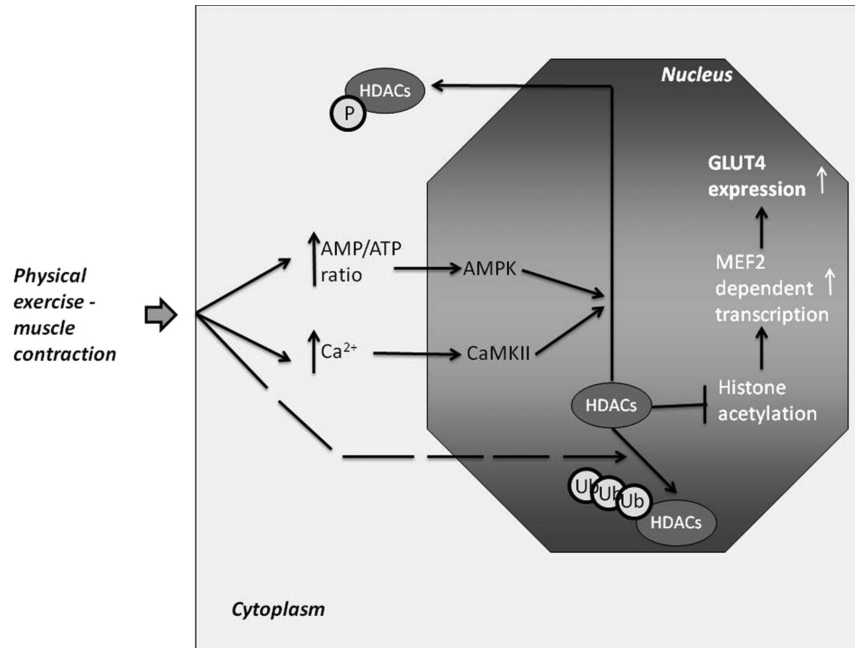


FIG. 2

Expression of GLUT4 expression by HDAC by exercise. This figure adapted from [6] with copyright permission.

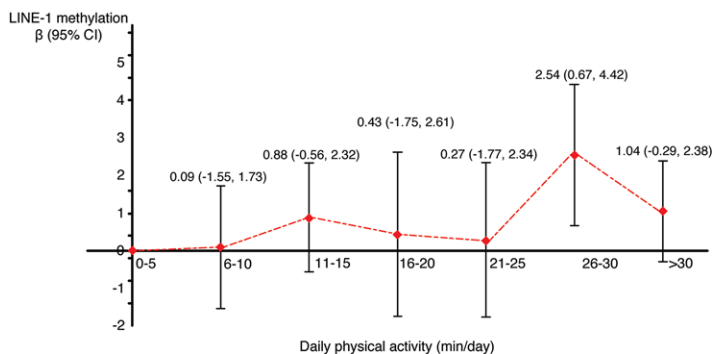


FIG. 3

The broad DNA methylation in adults. This figure adapted from [30] with copyright permission.

On the other hand, hypermethylation in the promoter regions of suppressor genes is one of the factors contributing to some types of cancer. Hypermethylation in tumor promoter regions is a suppressor of APC and RASSF1A genes. The suppression of these genes is an indication for cancer [32]. The role of the APC gene is the correct division of the cell and the control of the number of chromosomes after division, and the role of the RASSF1A gene is to interact with the XPA protein for DNA repair. Exercise has been shown to be a reducing agent, and even a suppressant of hypermethylation, as well as in reducing and even reversing the hypermethylation of APC and RASSF1A promoters, reducing their risk of cancer [32]. Environmental conditions, including appropriate physical pressure, strongly signal expression of the TP53 gene, which

causes epigenetic changes and suppresses cancerous tumors [33]. TP53 is a gene code for p53 protein, an important protein in the pathway of apoptosis, and also the p53 protein is important for regulating cell growth, so the hypermethylation of the TP53 promoter region is a common indicator of cancer development [33].

Research has also focused on the effects of exercise on breast cancer. One study found that exercise training for 6 months and a duration of 129 Min per round with moderate intensity reduced the methylation of three genes from 46 genes studied. For example, one of these genes was the suppressor gene of L3MBTL1 breast cancer tumors, which was investigated in this study.

In this protocol, the L3MBTL1 methylation training group decreased by 48.1%, while in the control group, the methylation of this gene increased by 15.2%. The findings of this study indicate that exercise will reduce the methylation of the suppressor gene of breast cancer tumors and thus have a positive effect on the improvement and prevention of this sample of cancer [24].

The above figure is a simple linear regression diagram that shows the relationship between physical activity and genomic DNA methylation. The horizontal axis of this chart shows the training groups (0–5, 6–10, 11–15, 16–20, 21–25, 26–30, >30 Min) with daily physically activities. As you can see, more methylation has been achieved in groups that have been on continuous training for 26–30 Min, compared with those who have been subjected to these activities for longer periods of time [30].

2.3. Sports and elderly

One of the important components of aging is the significant loss of DNA methylation over time [34, 35]. Methyl deoxy cytosine is involved in the process of differentiation and maintenance of

this distinction. Cellular differentiation is a process in which methylation from different regions occurs in DNA of a cell that can alter the transcription of genes. It has been shown that changes in histones, like DNA changes, affect the aging process [34]. Histone methylation, such as histone H3 di- or tri-methyl in the lysine sequence 3 or 4, can regulate gene expression changes, age-dependent genes, including HTERT and P16INK4 α , and thus increase the life span of human cells induced by CR share [34, 36].

In cell differentiation, DNA methylation is important for the identity and function of a cell due to its role in controlling gene expression. In a study that compared the DNA methylation of newborn babies and the elderly, results of this study showed that older people generally had a significant reduction in DNA methylation. With the onset of aging, the amount of DNA methylation begins to decrease slowly [37].

Studies have also been carried out in the methyl residue of doxy cytidine from organs of different ages of rodents. These studies have shown that loss of DNA methylation will significantly increase the aging process. Therefore, aging is associated with a significant loss of DNA methylation [34, 37]. However, this loss of DNA methylation seems to be reduced by exercise.

Studies have been done on the effect of exercise on DNA methylation and its effect on the aging process in humans. In general, exercise can act as an inhibitor in the aging process by reducing the rate of loss of DNA methylation [5].

Telomeres are one of the other causes of aging and burnout. The gradual shortening of the telomeres at the end of the chromosomes will affect the aging process. Aging and age-related diseases are associated with a significant shortening of these sequences. The collapse of telomeres occurs in somatic cells, in which telomerase (telomere length control enzyme) is expressed [38]. One study showed that exposed mice exhibited a significant reduction in telomere endurance compared to the control group [39]. Therefore, exercise as an effective factor in the reduction of telomere endurance and telomeric stability can slow down the aging process [39, 40].

In addition to the above, research has been conducted on the impact of exercise exercises in chronic COPD-related illnesses, most of which are those of the elderly. In some studies, it has been shown that the exacerbation of some respiratory diseases, such as COPD, may be due to the consequences of intense exercise [41, 42]. Several studies have shown that after 24 H of exercise and after 72 H of rest, DNA damage is observed [43]. In other studies, DNA has been shown to be damaged by healthy people [44]. In spite of these investigations, the molecular mechanisms related to these useful effects are yet to be fully explored. Latest evidences have shown that different protocols of exercise control epigenetic signals in peripheral blood from diverse populations [45–48]. However, as per our knowledge, no study has described the impact of exercise on epigenetic signals in people with COPD.

In most studies, it has been determined that DNA damage and oxidation factors occur due to inflammation [49]. In

patients with COPD, the antioxidant defense system is less resistant to reactive oxygen species (ROS) compared to healthy subjects, and this issue is due to increased oxidative stress response after exercise [50]. Cells of the human body are continuously producing ROS and free radicals due to their metabolic processes [51]. Under normal conditions, it is neutralized by an antioxidant defense system (endogenous) [52]. During extreme sports, the production of ROS and free radicals increases, leading to oxidative stress (damage to DNA, protein oxidation, lipid peroxidation) and muscle damage that produces muscular and reduces performance and speeds up fatigue [51, 53].

One of the factors causing oxidative damage is the production of uric acid produced by maximum exercise [54]. In a study, it was found that a moderate and submaximal exercise group would significantly decrease plasma uric acid [55]. In people with COPD, severe hypoxia will occur, due to the fact that they tend to have superficial and shallow breaths and therefore a greater number of tail and exhale, and these physiological factors will increase the respiratory dead space, resulting in an increase in load and pressure in the alumni. This reduction in oxygen eventually ends in inflammation [56]. This may be associated with an increase in hydrogen peroxide (H₂O₂) in these patients [57].

Exercise with moderate intensity will reduce the oxidative DNA damage in healthy people and patients with COPD. Moderate exercise can act as an antioxidant, because these exercises increase the expression of two types of enzymes called SOD and GPX [58]. Unlike the high and extreme exercises in COPD patients that increase oxidative stress, moderate and submaximal exercises reduce oxidative stress and complications [59].

On the other hand, the reduction of endurance of physical pressure and impairment in quality of life, which is largely due to dysfunction of the skeletal system, is a major complaint of patients with COPD. Evidence suggests that endurance training with an intensity of 60–70% of peak heart rate improves muscle oxygen transmission, decreases blood lactate levels, and increases the ability of oxidative mitochondria in moderate to severe COPD patients [59]. Musculoskeletal atrophy is one of the most common causes in COPD patients. In patients with COPD, one of the mortality factors, the underlying causes are the muscular and immobile impairment of these patients. This muscle weakness occurs moderately to severe due to the immobility and metabolic factors of COPD. Sports activities with changes in structure and muscle metabolism are one of the factors preventing the complications of these patients. One of these changes is the MiRNA variation, which plays a role in regulating the muscular structure [60], which is briefly explained in the previous studies [2, 6].

3. Conclusion

Changes in gene expression, which are mainly due to changes in the posttranslational role of histone proteins, will play a major role in metabolic and muscular changes. Various

studies have shown that sports activities cause posttranslation changes in histone proteins and DNA methylation in chromatin, which is the key to gene expression. In addition, most of the studies conducted in this field have a positive effect of regular exercises on the expression of histone proteins and DNA methylation and the prevention of certain diseases, such as cancer and diabetes, as well as the rehabilitation of patients with pulmonary obstruction. In the elderly, it is reported to result in an increase in the antioxidant activity of the cell. In this study, the statistical population is limited to a few specific categories, since epigenetics and gene expression are almost the cornerstone of all body interactions, especially regarding metabolic changes in the body through exercise. Therefore, more studies are needed in this regard.

The authors declare no conflict of interest.

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